

Evaluation Framework

JEPA RAG vs Transformer RAG — Metric Design & Execution Plan

TL;DR

- **Goal:** Prove JEPA-refined graphs produce better RAG than Transformer-refined graphs.
- **How:** Same RAG pipeline, swap in 3 graphs (No Refinement / Transformer / JEPA), compare.
- **16 metrics from 4 papers:**
 - RAGAS (Es et al., 2023) — automated RAG quality
 - MediGRAF (Thio et al., 2025) — expert evaluation + query complexity
 - Clin-JEPA (Yang et al., 2026) — latent geometry
 - LeCun (2022) — world model properties
- **Phase 1 — Automated (5 metrics):** Faithfulness, Answer Relevance, Context Relevance, Proxy Hallucination Rate, Recall/Precision/F1. Needs: query set + refined graphs + RAG pipeline. No ground-truth graph required. No expert required.
- **Phase 1 — Expert (5 metrics):** Accuracy, Completeness, Relevance & Conciseness, Overall Quality, Safety. Requires 2 clinical expert evaluators.
- **Phase 2 (after model training, 6 metrics):** Cohort Displacement Ratio, Cohen's d, UMAP, Sample Efficiency, Inference Speed, OOD Robustness. Needs: trained model embeddings.

Discussion Agenda (Jun 2):

1. Which metrics do we prioritize? (e.g., automated-only first, or automated + expert together?)
2. Do we commit to Phase 2, or scope to Phase 1 only?

1. Overview

1.1 Objective

This document defines the evaluation framework for the Evaluation workstream. The original L1 intrinsic metrics (Entity F1, Relation F1, Completeness) are replaced with **RAG-based metrics** to systematically verify whether JEPA-based graph refinement yields superior clinical RAG performance compared to Transformer-based refinement.

1.2 Key Hypothesis

	Transformer RAG	JEPA RAG
Search mechanism	Relies on surface-level semantic similarity matching	Leverages structural consistency in latent space
Weakness	May retrieve relations that are locally plausible but clinically inconsistent in noisy clinical notes	—

Transformer RAG		JEPA RAG
Strength	—	Energy-based plausibility scoring preserves only structurally coherent edges, improving retrieval quality

1.3 Experimental Conditions

All experiments hold the **RAG pipeline constant** and vary only the graph refinement method.

Condition	Graph Construction
(A) No Refinement	GLiNER2-extracted raw graph used directly
(B) Transformer Refinement	GLiNER2 raw graph → Transformer-based refinement
(C) JEPA Refinement	GLiNER2 raw graph → Graph-JEPA-based refinement

- **Data:** MIMIC-IV-Note (discharge summaries)
- **Controlled variables:** Embedding model, retriever (top-K), generator (LLM) — all held constant
- **Independent variable:** Graph refinement method (A, B, C)

1.4 Evaluation Structure

Evaluation is executed in two phases.

Phase	Scope	Prerequisites
Phase 1	RAG automated metrics + expert evaluation	Refined graphs + existing RAG pipeline
Phase 2	Latent geometry + world model properties	Trained models + embeddings

1.5 References

Paper	Citation	Metrics Adopted
Ragas: Automated Evaluation of Retrieval Augmented Generation	Es et al., 2023. arXiv:2309.15217	Faithfulness, Answer Relevance, Context Relevance
Clin-JEPA: A Multi-Phase Co-Training Framework for Joint-Embedding Predictive Pretraining on EHR Patient Trajectories	Yang et al., 2026. arXiv:2605.10840	Cohort Displacement Ratio, Cohen's d, UMAP visualization (three-axis evaluation framework)
MediGRAF: Medical Graph Retrieval Augmented Framework	Thio et al., 2025. arXiv:2602.00009	Query complexity classification, Likert-scale expert evaluation (Accuracy, Completeness, Relevance & Conciseness, Overall Quality, Safety), Recall / Precision / F1

Paper	Citation	Metrics Adopted
A Path Towards Autonomous Machine Intelligence	LeCun, 2022. Open Review	Sample Efficiency, Inference Speed, Robustness to OOD (world model evaluation axes)

2. Phase 1: RAG-Based Evaluation

Metrics that can be **executed immediately** with refined graphs and the existing RAG pipeline. No ground-truth graph required.

2.1 Automated Evaluation — RAGAS Framework

Automated scoring via LLM-as-judge using the RAGAS library. No manual annotation required.

① Faithfulness — *Es et al., 2023*

- **Definition:** "The answer is faithful to the context if the claims that are made in the answer can be inferred from the context." (Es et al., 2023)
- **Procedure:**
 1. Prepare a clinical query set (50–100 queries) and run RAG under conditions A, B, C
 2. "Given a question and answer, create one or more statements from each sentence in the given answer" — decompose the answer into a set of statements S
 3. For each statement s_i , "determine whether they are supported by the information present in the context" using a verification function $v(s_i, c(q))$
- **Formula:** $F = |V| / |S|$, where $|V|$ is the number of supported statements and $|S|$ is the total number of statements (Es et al., 2023)
- **Tool:** `from ragas.metrics import faithfulness`
- **Expected:** C (JEPA) > B (Transformer) > A (No Refinement)

② Answer Relevance — *Es et al., 2023*

- **Definition:** "The answer is relevant if it directly addresses the question in an appropriate way. The assessment of answer relevance does not take into account factuality, but penalises cases where the answer is incomplete or where it contains redundant information." (Es et al., 2023)
- **Procedure:** For the given answer $a_s(q)$, "the LLM generates n potential questions q_i " from the answer. For each q_i , cosine similarity with the original question q is computed using embeddings.
- **Formula:** $AR = (1/n) * \sum \text{sim}(q, q_i)$ (Es et al., 2023)
- **Tool:** `from ragas.metrics import answer_relevancy`

③ Context Relevance — *Es et al., 2023*

- **Definition:** "The context is considered relevant to the extent that it exclusively contains information that is needed to answer the question. This metric aims to penalise the inclusion of redundant information." (Es et al., 2023)
- **Procedure:** "The LLM extracts a subset of sentences, S_{ext} , from $c(q)$ that are crucial to answer q ." (Es et al., 2023)

- **Formula:** $CR = \text{number of extracted sentences} / \text{total number of sentences in } c(q)$ (Es et al., 2023)
- **Tool:** `from ragas.metrics import context_relevancy`
- **GT Required:** ✗
- **Significance:** JEPA-refined graphs remove noisy edges, so retrieved context should contain less irrelevant information

④ Proxy Hallucination Rate — *derived from RAGAS Faithfulness*

- **Definition:** The proportion of claims in the generated answer that are unsupported by the retrieved context. Note: this is not a direct measure of medical hallucination, but a proxy based on the rate of claims unsupported by retrieved context.
- **Formula:** $\text{Proxy Hallucination Rate} = 1 - \text{Faithfulness}$
- **Note:** Automatically derived from ④ Faithfulness. No separate experiment required.

2.2 Query Complexity Classification — *Thio et al., 2025*

Queries are classified into three complexity levels following MediGRAF (Thio et al., 2025), where "complexity is defined based on the graph traversal depth and data synthesis requirements."

Complexity	Definition (Thio et al., 2025)	Example	Metrics Applied
Simple	"Direct fact retrieval requiring single node lookups or straightforward relationships, such as patient demographics, admission counts, or document tallies."	"What is the patient's primary diagnosis?"	Recall, Precision, F1 (automated)
Medium	"Traversal of multiple relationships or temporal reasoning, including medication histories, diagnostic patterns across admissions, or laboratory result trends."	"List all antibiotics prescribed during the pneumonia admission."	Recall, Precision, F1 (automated)
Complex	"Synthesis, summarisation, or inference from multiple data sources, such as comprehensive patient summaries, differential diagnosis reasoning, or treatment response analysis."	"Summarise the patient's response to antibiotic treatment across all admissions."	Likert-scale expert evaluation

- **Simple / Medium:** MIMIC structured data (diagnosis codes, prescription records) serve as approximate ground truth for automated Recall, Precision, F1 computation. Note: structured labels may not exactly match free-text expressions in discharge notes (e.g., ICD code "F31.1" vs. narrative "bipolar disorder, type 1"), so these are treated as approximate rather than exact ground truth.
- **Complex:** Free-text responses without discrete ground-truth answers; evaluated via expert assessment

2.3 Expert Evaluation — *Thio et al., 2025*

For complex queries, clinical experts evaluate responses using 5-point Likert scales following MediGRAF (Thio et al., 2025). "Two independent clinical reviewers assessed responses against a gold-standard summary created by a senior clinician."

Dimension	Definition (Thio et al., 2025)	Scale
Accuracy	"The factual correctness of the generated narrative. A score of 5 indicates all medical facts (dates, dosages, diagnoses) are correct compared to the source notes."	1–5
Completeness	"Whether the answer provides all key pieces of information present in the ground truth."	1–5
Relevance & Conciseness	"Whether the answer directly addresses the question without redundant or off-topic information."	1–5
Overall Quality	"A holistic judgment of the answer's clinical utility."	1–5
Safety	"A binary metric where '1' (Unsafe) indicates the presence of a critical hallucination: inventing a medical condition, hallucinating a medication not in the record, or contradicting a known contraindication."	Binary (0/1)

- **Evaluators:** 2 clinical experts
- **Evaluation targets:** Complex query responses across conditions A, B, C
- **Reference:** Expert-created gold-standard summaries for comparison (following MediGRAF: "two independent clinical reviewers assessed responses against a gold-standard summary created by a senior clinician")
- **Scoring rubric:** Adapted from MediGRAF Appendix B annotation manual

2.4 Phase 1 Summary

#	Metric	Type	GT Required	Tool	Source
①	Faithfulness	Automated	✗	RAGAS	Es et al., 2023
②	Answer Relevance	Automated	✗	RAGAS	Es et al., 2023
③	Context Relevance	Automated	✗	RAGAS	Es et al., 2023
④	Proxy Hallucination Rate	Automated	✗	1 - ①	Derived from RAGAS
⑤	Recall / Precision / F1 (Simple, Medium)	Automated	Δ (approximate)	Custom	Thio et al., 2025
⑥	Accuracy (Likert)	Manual	Δ (expert-created reference)	Expert	Thio et al., 2025
⑦	Completeness (Likert)	Manual	Δ (expert-created reference)	Expert	Thio et al., 2025

#	Metric	Type	GT Required	Tool	Source
⑧	Relevance & Conciseness (Likert)	Manual	Δ (expert-created reference)	Expert	Thio et al., 2025
⑨	Overall Quality (Likert)	Manual	Δ (expert-created reference)	Expert	Thio et al., 2025
⑩	Safety (Binary)	Manual	✗	Expert	Thio et al., 2025

2.5 Phase 1 Action Items

If members agree, then:

Item
Prepare query set (Simple / Medium / Complex, 50–100 queries)
Set up RAGAS environment (<code>pip install ragas</code>)
Receive refined graphs × 3 (A, B, C)
Execute RAG pipeline with each graph and collect results
Recruit 2 clinical experts (maybe Amit's network?)
Execute Likert-scale evaluation
Compile results tables and figures

2.6 Expected Results Tables

Automated Evaluation (RAGAS)

Metric	No Refinement (A)	Transformer (B)	JEPA (C)
Faithfulness ↑			
Answer Relevance ↑			
Context Relevance ↑			
Proxy Hallucination Rate ↓			

Recall by Query Complexity

Complexity	No Refinement (A)	Transformer (B)	JEPA (C)
Simple			
Medium			
Complex	N/A (Likert)	N/A (Likert)	N/A (Likert)

Expert Evaluation (Likert, Complex Queries)

Dimension	No Refinement (A)	Transformer (B)	JEPA (C)
Accuracy (1–5)			
Completeness (1–5)			
Relevance & Conciseness (1–5)			
Overall Quality (1–5)			
Safety Violations	/N	/N	/N

3. Phase 2: Graph/Model-Dependent Evaluation

Metrics that require **trained models and embeddings** as prerequisites. Execution begins upon receipt of model outputs.

3.1 Latent Geometry — *Yang et al., 2026*

Objective: Verify whether Graph-JEPA learns a "clinically discriminative latent geometry" (Yang et al., 2026) that captures meaningful patient-state structure.

Input: Per-patient graph embedding vectors extracted from the trained model.

MIMIC note → GLiNER2 → raw graph → Refinement (A/B/C) → graph embedding vector

⑪ Cohort Displacement Ratio — *Yang et al., 2026*

- **Definition:** The ratio of centroid displacement between a deteriorating cohort and a stable cohort in latent space. "The diagnostic of a clinically grounded encoder is not how far deteriorating patients move, but how steady stable patients are kept." (Yang et al., 2026)
- **Procedure:**
 1. Select two cohorts from MIMIC: Deteriorating ($\Delta\text{SOFA} \geq 3$ or `hospital_expire_flag=1`) vs. Stable ($\text{SOFA std} \leq 1$ or discharged home)
 2. Extract per-patient graph embeddings under conditions A, B, C
 3. Compare centroid displacement distance between the two cohorts
- **Formula:** $\text{Displacement Ratio} = \frac{\text{deteriorating cohort centroid displacement}}{\text{stable cohort centroid displacement}}$
- **Reference values:** "4.83x for CLIN-JEPA, 2.62x for V-JEPA 2-AC style, and 1.03x for SFT" (Yang et al., 2026)

⑫ Cohen's d + Statistical Significance — *Yang et al., 2026*

- **Definition:** Effect size and statistical significance of cohort displacement separation
- **Procedure:**
 1. Compute per-patient net displacement (start → final embedding distance)
 2. Compare Deteriorating vs. Stable groups via Cohen's d and one-sided Mann-Whitney U test

- **Threshold:** Cohen's $d \geq 0.5$ (medium effect), $p < 0.001$
- **Reference:** "Per-patient net-displacement separation reaches Cohen's $d=0.598$ (medium-large effect, $p=2.7 \times 10^{-4}$) for CLIN-JEPA, versus $d=0.058$ (n.s., $p=0.53$) for V-JEPA 2-AC style" (Yang et al., 2026)
- **Tool:** `scipy.stats.mannwhitneyu`

⑬ UMAP Visualization — Yang et al., 2026

- **Definition:** 2D projection of patient embeddings showing cohort separation across conditions
- **Procedure:** Per-condition UMAP projection → Deteriorating (red) vs. Stable (blue), three-panel layout
- **Reference:** Adapted from Clin-JEPA Figure 3: "per-encoder trajectories with cohort means highlighted along a time-coded color band" (Yang et al., 2026)
- **Tool:** `umap-learn`, `matplotlib`

3.2 World Model Properties — LeCun, 2022

Objective: Verify whether JEPA possesses fundamental advantages over Transformers as a world model.

⑭ Sample Efficiency — LeCun, 2022

- **Definition:** The amount of training data required to achieve equivalent performance
- **Procedure:**
 1. Train refinement models using 1%, 5%, 10%, 50%, 100% of the MIMIC cohort
 2. Run RAG with each model's refined graph → measure Faithfulness
 3. Plot learning curves: X-axis (data fraction) × Y-axis (Faithfulness)
- **Prerequisites:** 5 data fractions × 2 model types (JEPA, Transformer) = 10 model training runs
- **Difficulty:** High

⑮ Inference Speed — LeCun, 2022

- **Definition:** End-to-end processing time comparison for an identical query set
- **Procedure:** Measure latency across three stages independently:
 1. Graph Refinement time
 2. Retrieval time
 3. Generation time
- **Prerequisites:** Trained models in inference-ready state
- **Difficulty:** Low

⑯ OOD Robustness — LeCun, 2022

- **Definition:** Degree of performance degradation under input noise
- **Procedure:**
 1. Measure baseline Faithfulness on original MIMIC notes
 2. Apply artificial corruption at three levels (10%, 20%, 30%): abbreviation variants, irrelevant sentence insertion, sentence order shuffling
 3. Re-run pipeline on corrupted notes → re-measure Faithfulness
- **Formula:** `Robustness Score = corrupted Faithfulness / clean Faithfulness`

- **Difficulty:** Medium

3.3 Phase 2 Summary

#	Metric	Type	Prerequisites	Source
⑪	Cohort Displacement Ratio	Automated	Trained model embeddings	Yang et al., 2026
⑫	Cohen's d	Automated	Derived from ⑪	Yang et al., 2026
⑬	UMAP Visualization	Automated	Derived from ⑪	Yang et al., 2026
⑭	Sample Efficiency	Automated	Models ×10	LeCun, 2022
⑮	Inference Speed	Automated	Model inference	LeCun, 2022
⑯	OOD Robustness	Automated	Corruption functions + models	LeCun, 2022

3.4 Expected Results Tables

Latent Geometry

Based on Clin-JEPA (Yang et al., 2026), where JEPA achieved 4.83×, V-JEPA 2-AC 2.62×, and SFT baseline 1.03×, we expect a similar trend in our graph refinement setting:

Metric	No Refinement (A)	Transformer (B)	JEPA (C)
Displacement Ratio ↑			
Cohen's d ↑			
p-value			

World Model Properties

Metric	Transformer (B)	JEPA (C)
10% data → Faithfulness		
Inference Speed (sec) ↓		
OOD Robustness (30% corruption → drop%) ↓		

Appendix: Expert Evaluation Scoring Rubric

Adapted from MediGRAF Appendix B (Thio et al., 2025).

Accuracy

- **5 (Very Good):** "The model's answer is completely factually correct and aligns perfectly with the ground truth. No errors."
- **4 (Good):** "The model's answer is mostly accurate with only minor, insignificant inaccuracies that do not mislead."

- **3 (Fair):** "The model's answer contains some noticeable inaccuracies, but the main point might still be partially correct or understandable."
- **2 (Poor):** "The model's answer contains significant factual errors that make it misleading or incorrect."
- **1 (Very Poor):** "The model's answer is completely factually incorrect or fabricated."

Completeness

- **5 (Very Good):** "The model's answer includes all relevant information present in the ground truth; it is fully comprehensive."
- **4 (Good):** "The model's answer includes most of the relevant information, with only minor omissions that don't critically affect the answer's utility."
- **3 (Fair):** "The model's answer provides some relevant information but omits one or more key pieces of information found in the ground truth."
- **2 (Poor):** "The model's answer omits significant and critical pieces of information, making it substantially incomplete."
- **1 (Very Poor):** "The model's answer provides very little or none of the relevant information present in the ground truth."

Relevance & Conciseness

- **5 (Very Good):** "The model's answer is perfectly focused on the question, concise, and contains no irrelevant information."
- **4 (Good):** "The model's answer is mostly relevant and concise, with only minimal extraneous information that is easily ignorable."
- **3 (Fair):** "The model's answer addresses the question but includes some noticeable irrelevant or redundant information, or is somewhat verbose."
- **2 (Poor):** "The model's answer contains significant irrelevant information, or largely misses the point of the question, or is overly verbose."
- **1 (Very Poor):** "The model's answer is completely irrelevant, off-topic, or excessively verbose to the point of being unhelpful."

Overall Quality

- **5 (Very Good):** "Excellent answer. Accurate, complete, relevant, and clear. Fully usable."
- **4 (Good):** "Good answer. Minor issues in one area but generally accurate, complete, and useful."
- **3 (Fair):** "Acceptable answer. Some issues with accuracy, completeness, or relevance, but still provides some value."
- **2 (Poor):** "Poor answer. Significant issues make it largely unusable or misleading."
- **1 (Very Poor):** "Unacceptable answer. Completely incorrect, incomplete, or irrelevant."

Safety

- **0 (Safe):** "Answer was clinically safe and would not have resulted in patient harm."
- **1 (Unsafe):** "Answer was clinically unsafe and would have resulted in missed information and patient harm."

Following MediGRAF (Thio et al., 2025): "A score of 1 (Unsafe) indicates the presence of a critical hallucination: inventing a medical condition, hallucinating a medication not in the record, or contradicting

a known contraindication."